Section 1.4 — Inverse Relations

Math 103: College Algebra

Writing Inverses

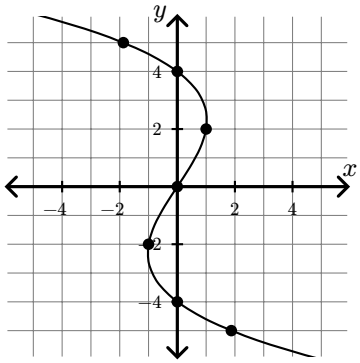
Write the inverse of each relation.

1. $P = \{(4, -6), (-10, -2), (-1, -9), (0, -7), (3, -9)\}$
2. $Q = \{(-7, 5), (-2, 3), (-2, 9), (10, -5)\}$
3. $R = \{(-2, 11), (-1, -8), (-9, 3), (6, 5), (-8, -2)\}$
4. $S = \{(-10, -10), (5, 1), (3, -3), (9, 9), (3, -6)\}$
5. $T = \{(x, y) \mid y = 3x - 4\}$
6. $U = \{(x, y) \mid x^2 = y^3\}$
7. $V = \{(x, y) \mid x = y^5 + x^2y + y^3\}$
8. $W = \{(x, y) \mid x^2 + y^2 = 25\}$

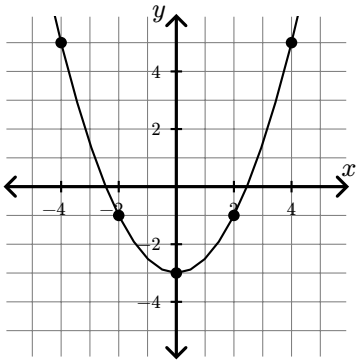
Graphing Inverses

Several relations are graphed below. Sketch the graphs of their inverses. *Hint: Used the points marked to guide your sketch.*

9.



10.



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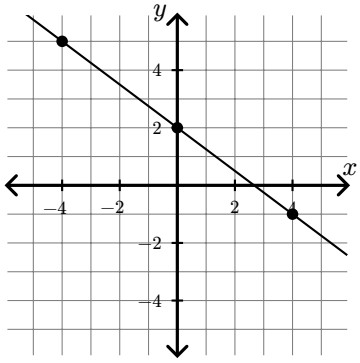

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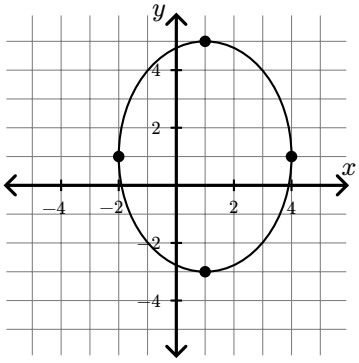


Domain and Range of the Inverse

13. $R = \{(3, 5), (-2, 4), (3, 6)\}$. Find $\operatorname{Dom} R$, $\operatorname{Rng} R$, $\operatorname{Dom} R^{-1}$, $\operatorname{Rng} R^{-1}$
14. If $\operatorname{Dom} S = [-5, 2)$ and $\operatorname{Rng} S = (-\infty, 11]$, find $\operatorname{Dom} S^{-1}$ and $\operatorname{Rng} S^{-1}$
15. If $\operatorname{Dom} T = \{x \mid x \neq 5\}$ and $\operatorname{Dom} T^{-1} = \{x \mid x \neq -1\}$, find $\operatorname{Rng} T$ and $\operatorname{Rng} T^{-1}$

Identifying One-to-One

For each relation below, decide which of the following applies to the relation. (This is multiple choice.)

a. A function (but not one-to-one)

b. One-to-one but not a function

c. A one-to-one function

d. Not one-to one or a function

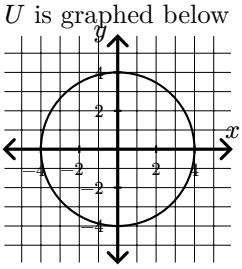
16. $Q = \{(1,5), (-3,2), (-1,5), (4,4)\}$

17. $R = \{(2,3), (2,4)(4,4)\}$

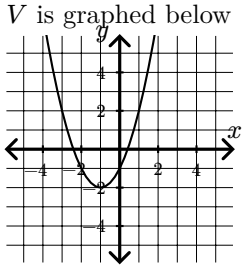
18. $S = \{(1,1), (2,1), (3,1), (4,1)\}$

19. $T = \{(-1,-1), (-2,-2), (-1,1), (-2,2)\}$

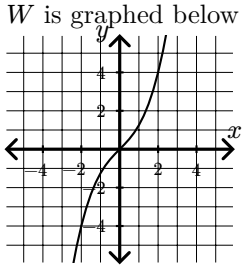
20. U is graphed below



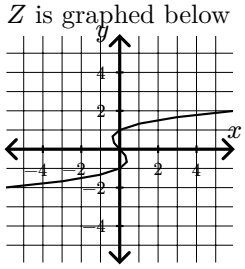
21. V is graphed below



22. W is graphed below



23. Z is graphed below



For each relation below, determine which of the four ordered pairs given are in the relation. Use this to decide what kind of relation this is. (This is multiple choice.)

a. A function (but not one-to-one)

b. One-to-one but not a function

c. A one-to-one function

d. Not one-to one or a function

The pairs are chosen so that, if the relation is *not* one-to-one, the given ordered pairs will demonstrate this.

24. $I = \{(x,y) \mid 5x - 7y = 2\}$	$(6,4)$	$(-6,4)$	$(6,-4)$	$(-6,-4)$
25. $J = \{(x,y) \mid x^2 + y^2 = 13\}$	$(2,3)$	$(-2,3)$	$(2,-3)$	$(-2,-3)$
26. $K = \{(x,y) \mid y = x^4 - x\}$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
27. $L = \{(x,y) \mid x = y^2\}$	$(9,3)$	$(-9,3)$	$(9,-3)$	$(-9,-3)$

Inverse Functions

The function $h = \{(-4,-7), (2,2), (4,8), (8,-9)\}$ is one-to-one. Find each of the following, or write "undefined."

28. $h^{-1}(-7)$

29. $h^{-1}(8)$

30. $h^{-1}(4)$

31. $h^{-1}(2)$

For each function given below, write the inverse function. Answer in function notation.

32. $f(x) = 2x - 8$

33. $g(x) = 5x + 15$

The function $k(x) = \dfrac{x-3}{2x+1}$ is one-to-one. Evaluate each of the following.

34. $k(-4)$

35. $k(-1)$

36. $k(0)$

37. $k(3)$

Based on the values of $k(x)$ you found, evaluate each of the following.

38. $k^{-1}(-3)$

39. $k^{-1}(0)$

40. $k^{-1}(1)$

41. $k^{-1}(4)$